

STUDY-RL

A 24-week self-paced curriculum in Deep Reinforcement Learning

Foundations · Deep RL · LLM RL
Safety-critical RL · Matching applications

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Preface

This document is the master syllabus for STUDY-RL, a 24-week self-paced curriculum designed for a learner who is fluent in modern deep learning but new to reinforcement learning. An additional applied audit in week 25 connects the core material to a synthetic ATS experiment and an application review. It threads three goals through every week:

1. **Strong applied mathematics** — every algorithm derived from first principles with intuition and code.
2. **State-of-the-art content** — modern best-practice as of mid-2026 (DPO/GRPO/RLVR, DreamerV3, EfficientZero v2, BBF, CrossQ, diffusion policies, LLM-as-agent...).
3. **Two applied threads** — *safety-critical RL* (one callout per lecture plus a dedicated 3-week track) and *matching/ranking* (insurance lead matching and recruitment candidate matching, woven through bandits, OPE, fairness).

Each week includes a teaching guide (`teaching_guide.md`), a Tufte-flavored PDF (`notes.pdf`), a Jupyter workbook with graded exercises (`workbook.ipynb`), curated extended reading (`extended_reading.pdf`), and a mini-project with tests (`project/`, `tests/`).

How to use this curriculum

- About **10 hours per week**: 2–3 h reading the PDF, 3–4 h on the workbook, 3–4 h on the project.
- Start with `STUDY_GUIDE.md` in the repository root, then the weekly teaching guide. Predict an update, calculate it by hand, and explain the result before running it.
- Attempt workbook exercises before opening hints or reference solutions. Some supplied cells call the reference directly: these demonstrate a solution but do not test an independent implementation.
- Check both execution and understanding: reproduce a numerical example, explain the assumptions, test a changed input, and compare a project result with a baseline. Passing tests alone does not establish mastery.

1 Curriculum overview

1.1 Tracks

- T1 — Foundations** (Weeks 1–4) Bandits, MDPs, Bellman equations, dynamic programming, Monte Carlo, TD/SARSA/Q-learning.
- T2 — Deep value-based** (Weeks 5–6) DQN, Double/Dueling/PER, distributional RL, Rainbow, BBF.
- T3 — Policy methods** (Weeks 7–10) REINFORCE, A2C, GAE, TRPO, PPO, SAC/TD3, CrossQ.
- T4 — Bandits & matching (applied)** (Weeks 11–12) LinUCB/TS, OPE, CLTR, fairness, two-sided markets.
- T5 — Multi-agent RL** (Week 13) MADDPG, MAPPO, self-play, PSRO, AlphaZero case study.
- T6 — Offline RL** (Weeks 14–15) CQL, IQL, AWR, TD3+BC, Decision Transformer, diffusion policies.
- T7 — Model-based RL** (Week 16) Dreamer V3, MuZero/EfficientZero V2, TD-MPC2, DIAMOND.
- T8 — Exploration & meta-RL** (Week 17) RND, ICM, BYOL-Explore, MAML/PEARL, in-context RL.
- T9 — LLM RL** (Weeks 18–21) RLHF/PPO, DPO/IPO/KTO/ORPO/SimPO, GRPO/RLOO/R-LVR, PRMs and reasoning RL.
- T10 — Safety-critical RL** (Weeks 22–24) CMDPs, CPO, shielding, formal verification, scalable oversight, capstone.

1.2 Week-by-week schedule

The schedule below describes topic coverage and suggested larger experiments. Several bundled mini-projects use smaller synthetic environments or reduced models; each weekly teaching guide identifies the implementation actually present and its limitations. A successful toy run is not a reproduction of a paper’s benchmark result.

Week	Topic	Mini-project
01	MDPs, bandits, the RL framing	Multi-armed bandit testbed + UCB/TS
02	Bellman equations, dynamic programming	Value/policy iteration on GridWorld
03	Monte Carlo prediction & control	Blackjack on-policy MC
04	TD, SARSA, Q-learning, Expected SARSA	CliffWalking + Taxi-v3

Week	Topic	Mini-project
05	Function approximation & Deep Q-Networks	DQN on LunarLander
06	Rainbow DQN + modern tricks (BBF, NoisyNets)	Rainbow on Atari (1 game, 100k steps)
07	Policy gradients & REINFORCE	REINFORCE + baselines on Cart-Pole
08	A2C, GAE, natural PG, TRPO	A2C+GAE on LunarLander-Continuous
09	Proximal Policy Optimization (deep dive)	PPO on BipedalWalker + the 37 details checklist
10	SAC, TD3, CrossQ for continuous control	SAC on MuJoCo Hopper
11	Bandits + contextual bandits + OPE	Off-policy evaluation of a synthetic insurance lead-market matcher
12	RL for ranking / matching (counterfactual LTR, fairness, two-sided)	ATS-style candidate-job ranker with fairness audit
13	Multi-agent RL + AlphaZero case study	Tennis self-play + small AlphaZero on Connect-Four
14	Offline RL fundamentals (CQL, IQL, AWR, TD3+BC)	IQL on D4RL halfcheetah-medium-replay
15	Decision Transformer + diffusion policies	DT on D4RL hopper-medium + a diffusion-policy toy task
16	Model-based RL (Dreamer V3, MuZero, TD-MPC2)	DreamerV3-lite on DM-Control walker-walk
17	Exploration + meta-RL + in-context RL	RND + MAML toy + Decision-Pretrained Transformer
18	RLHF + PPO for LLMs (from scratch and via TRL)	Summarization PPO on a Pythia-160M with LoRA
19	DPO + IPO + KTO + ORPO + SimPO	DPO vs. KTO vs. SimPO bake-off on Qwen-0.5B + UltraFeedback
20	GRPO + RLOO + RLVR + process reward models	GRPO with verifiable rewards on GSM8K-style math
21	Reward hacking, length bias, KL control, scalable oversight	Reward-hack lab + Constitutional-AI mini-loop
22	Constrained MDPs, Lagrangian methods, PPO-Lagrangian, CPO	Safety-Gymnasium PointGoal1 with PPO-Lagrangian
23	Safe exploration, shielding, robust RL, formal methods	LTL-shielded gridworld + distributionally robust DQN
24	Capstone: safe candidate-matching policy	End-to-end: offline-pretrained → constrained online → audited

1.3 Prerequisites

Fluent PyTorch, comfortable with probability, linear algebra, basic optimization theory. You do *not* need prior RL experience.

2 Bibliography pointers

The per-week `extended_reading.pdf` carries the focused paper list for that lecture. A consolidated bibliography is maintained at `research/`.